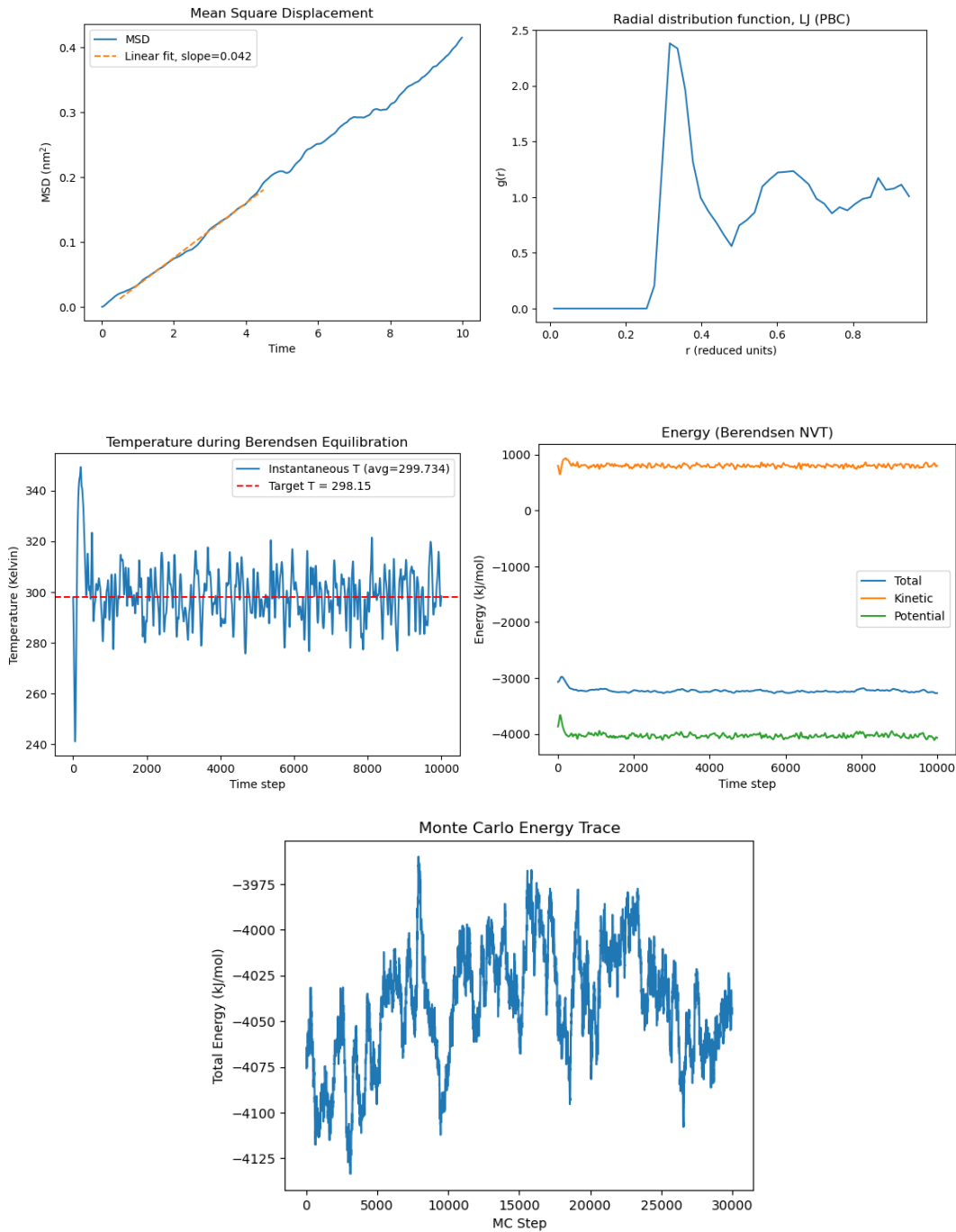


Appendix and Reference Molecular Simulation

Appendix

Results from the main code

Diffusivity = $7.0196 \times 10^{-9} \text{ m}^2/\text{s}$ at 298K



SAFT parameters

Table 2. Mie potential parameters used for the Molecular Dynamics model

	λ	ϵ/k_B (K)	σ (nm)	Source
Water	8.395	378.87	0.2915	(Herdes, Totton and Müller, 2015)
Propylene Oxide	11.877	228.679	0.308	Bottled SAFT, using critical conditions input

Mie potential

$$\phi(r) = \left(\frac{\lambda}{\lambda - 6} \right) \left(\frac{\lambda}{6} \right)^{6/(\lambda-6)} \varepsilon \left[\left(\frac{\sigma}{r} \right)^\lambda - \left(\frac{\sigma}{r} \right)^6 \right]$$

Calculating Mie potential parameters for Water and Pure PO

The Mie potential parameters (λ , ε , σ) were obtained using SAFT which required the input of critical temperature T_c , pressure P_c , critical density ρ_c , acentric factor, and liquid density at $0.7 \cdot T_c$, which reproduce the experimental data for the fluid. The critical parameters and acentric factor were obtained from Knovel's material property search as shown in Table 3. The unit of critical density of both PO and water were converted from g/ml to mol/m³ using molar mass of the molecules (18.015 g/mol for water, 58.08 g/mol for PO)

Table 3. Critical parameters used to calculate Mie potential parameters via SAFT-γ Mie, bottled

	T_c (K)	P_c (bar)	ρ_c (mol/m ³)	Acentric factor	Liquid density at $0.7 \cdot T_c$ (mol/m ³)	Source
Water	647.13	220.55	17900	0.3449	47990	(Yaws, 2014)
PO	482.25	49.20	5377	0.268	13292	(Yaws, 2010)

The Mie potential parameters calculated for both water and PO calculated using critical parameters and SAFT is shown in Figure 1 and Figure 2. It was decided not to use these parameters for water, and use the parameters from a literature (Herdes, Totton and Müller, 2015) because the epsilon value from the literature is higher, demonstrating smaller segment, and therefore more accuracy.

Segment 3 was chosen for PO, to balance between accuracy and computational running time of having small segments, this is chosen as PO is a complex molecule that is longer much longer than water.

The values you entered:

name	critical density (mol/m ³)	critical temperature (K)	critical pressure (bar)	acentric factor	Liquid density @ $T = 0.7 T_c$ (mol/m ³)
unnamed	17900	647.13	220.55	0.3449	47990

The parameters available:

# segments	lambda_r	lambda_a	epsilon (K)	sigma (m)
2	23.382219	6	519.152406	2.389950e-10
3	14.131349	6	348.250813	2.025470e-10
4	11.501065	6	269.033734	1.804260e-10
5	10.046364	6	220.163381	1.645540e-10
6	9.099352	6	186.481733	1.522920e-10

Figure 1. Water Mie potential parameters calculated from SAFT-γ Mie, bottled

The values you entered:

name	critical density (mol/m ³)	critical temperature (K)	critical pressure (bar)	acentric factor	Liquid density @ $T = 0.7 T_c$ (mol/m ³)
unnamed	5377	482.25	49.2	0.268	13292

The parameters available:

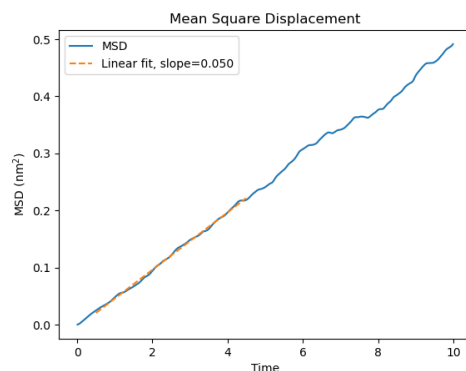
# segments	lambda_r	lambda_a	epsilon (K)	sigma (m)
2	17.079168	6	336.959666	3.618880e-10
3	11.876791	6	228.679203	3.084250e-10
4	10.026894	6	177.200792	2.747890e-10
5	8.937647	6	144.904248	2.504230e-10
6	8.217379	6	122.753018	2.316680e-10

Figure 2. Propylene oxide Mie potential parameters calculated from SAFT-γ Mie, bottled

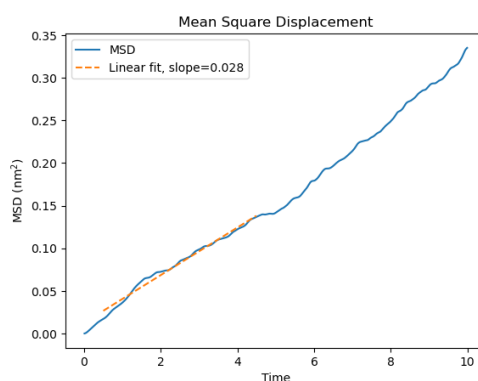
This significantly reduced computational cost and running time, while maintaining the important average energy levels and molecular characteristics of each species by integrating out the less important factors such as the atomic bonding and angles, and bond vibrations.

Additional results for predictive capacity

$T = 350\text{K}$ $D = 8.390 \times 10^{-9} \text{ m}^2/\text{s}$



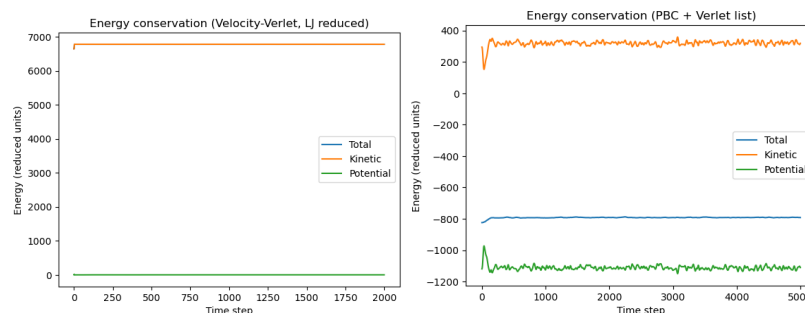
$T = 240\text{K}$ $D = 4.656 \times 10^{-9} \text{ m}^2/\text{s}$



Result for model progress from pure argon, pure water, pure PO and water-PO mixture

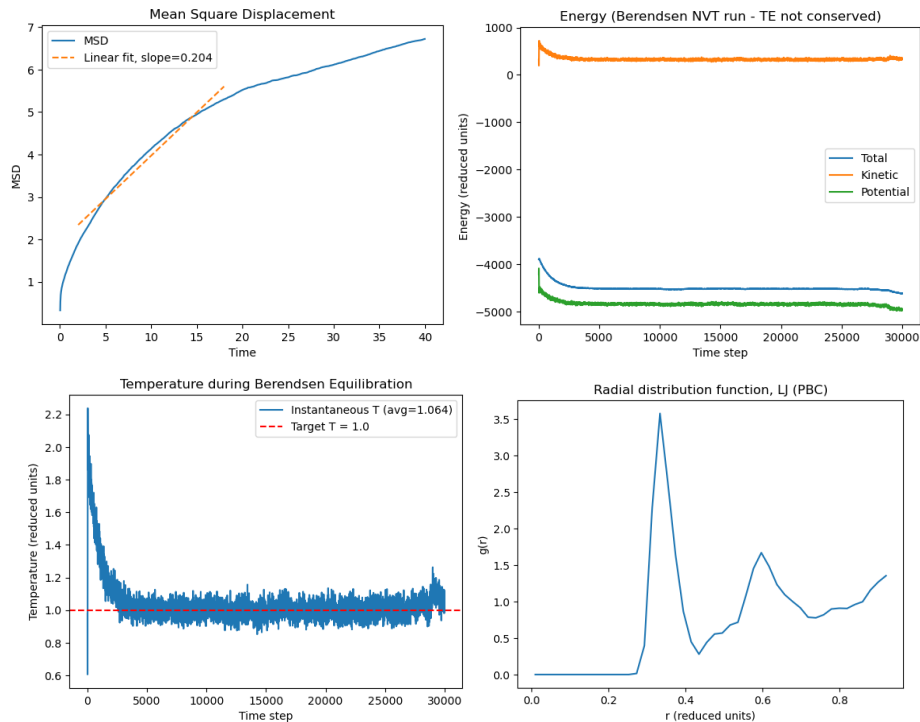
It has to be noted that the models below were calculated incorrectly, as the kinetic energy was still in the reduced unit in the results shown below. Therefore, the results of approximately $4 \times 10^{-9} \text{ m}^2/\text{s}$ for all the results are not practically correct, but here to show the progress of the model development. The model used for the result in the main report is in the correct GROMAC units as both kinetic and potential energy were calculated to be in kJ/mol.

Pure Argon model with reduced units, no PBC → with PBC

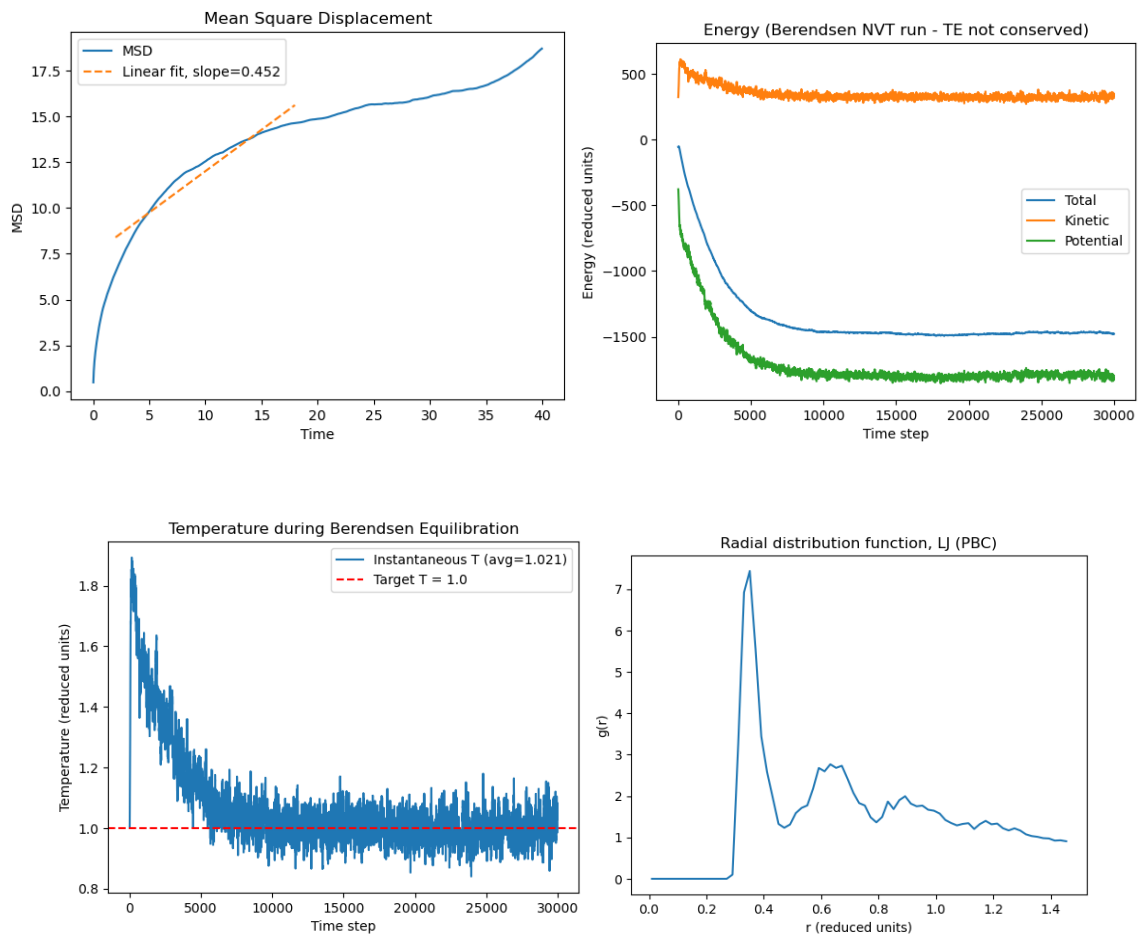


Pure water model with Mie potential parameters obtained from literature

$$D = 4.144 \times 10^{-9} \text{ m}^2/\text{s}$$

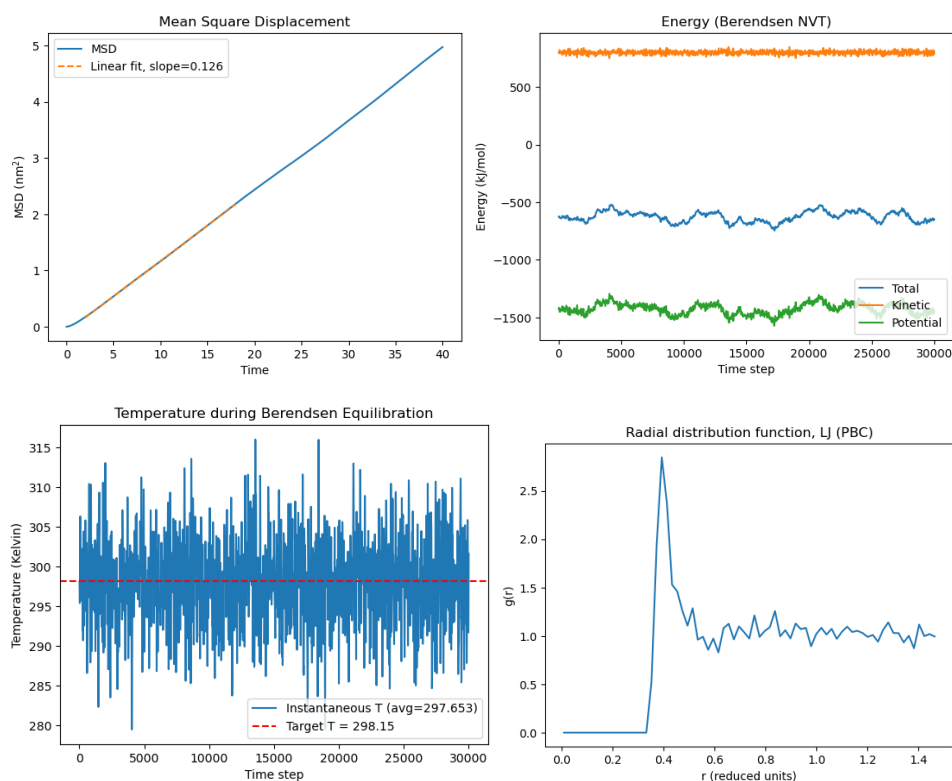
Pure Propylene oxide model with Mie potential parameters obtained from SAFT- γ Mie, bottled

$$D = 4.207 \times 10^{-9} \text{ m}^2/\text{s}$$



Fixed result of pure PO with all energy units in kJ/mol

$$D = 2.102 \times 10^{-8} \text{ m}^2/\text{s}$$



Reference

COMSOL, 2017. *Diffusion Coefficient Definition* [Online]. Available from: <https://www.comsol.com/multiphysics/diffusion-coefficient> [Accessed 10 December 2025].

Easteal, A.J., Price, W.E. and Woolf, L.A., 1989. Diaphragm cell for high-temperature diffusion measurements. Tracer Diffusion coefficients for water to 363 K. *Journal of the Chemical Society, Faraday Transactions 1: Physical Chemistry in Condensed Phases* [Online], 85(5), p.1091. Available from: <https://doi.org/10.1039/f19898501091>.

Herdes, C., Totton, T.S. and Müller, E.A., 2015. Coarse grained force field for the molecular simulation of natural gases and condensates. *Fluid Phase Equilibria* [Online], 406, pp.91–100. Available from: <https://doi.org/10.1016/j.fluid.2015.07.014>.

The Engineering ToolBox, 2003. *Water Density, Specific Weight and Thermal Expansion Coefficients - Temperature and Pressure Dependence* [Online]. Available from: https://www.engineeringtoolbox.com/water-density-specific-weight-d_595.html [Accessed 3 December 2025].

The Engineering ToolBox, 2004. *Specific Heat Capacity of Water: Temperature-Dependent Data and Calculator* [Online]. Available from: https://www.engineeringtoolbox.com/specific-heat-capacity-water-d_660.html?vA=298&units=K# [Accessed 11 December 2025].

Yaws, C.L., 2010a. *Knovel - Yaws' Thermophysical Properties of Chemicals and Hydrocarbons (Electronic Edition) - Critical Properties and Acentric Factors - Organic Compounds* [Online]. Available from: https://app.knovel.com/kn/resources/kt0078OO11/kpYTPCHE02/itable/itable?b-q=material%20or%20substance%20name%3Apropylene%2520oxide+AND+critical%20pressure%20mf%3A%5B+*+TO+*+%5D+AND+critical%20temperature%20mf%3A%5B+*+TO+*+%5D+AND+acentric%20factor%20mf%3A%5B+*+TO+*+%5D&columns=10101010%2C2%2C1%2C3%2C4%2C5%2C6%2C7%2C8%2C9%2C10%2C11%2C12%2C13%2C14%2C15%2C16%2C17%2C18&data_search_query=%3Fmn%3Dpropyle

[ne+oxide%3Aand%3A%5B+*+TO+*+%5D+AND+critical_pressure_mf%3A%5B+*+TO+*+%5D+AND+acentric_factor_mf%3A%5B+*+TO+*+%5D&sort_on=default](https://app.knovel.com/kn/resources/kt0078W7J1/kpYTPCHEE8/epgrph/equation?curve_ids=kr07662_YR5) [Accessed 3 December 2025].

Yaws, C.L., 2010b. *Knovel- 1,2-propylene oxide Diffusion Coefficients in Water at Infinite Dilution - Organic Compounds from Yaws' Transport Properties of Chemicals and Hydrocarbons (Electronic Edition)* [Online]. Available from: https://app.knovel.com/kn/resources/kt0078W7J1/kpYTPCHEE8/epgrph/equation?curve_ids=kr07662_YR5 [Accessed 5 December 2025].

Yaws, C.L., 2014. *Knovel - Yaws' Critical Property Data for Chemical Engineers and Chemists - Table 9. Critical Properties and Acentric Factor - Inorganic Compounds* [Online]. Available from: [https://app-knovel-com.ezproxy1.bath.ac.uk/kn/resources/kt009ZN2P4/kpYCPDCECD/itable/itable?b-q=material or substance name%3A+H2O+AND+critical temperature mf%3A%5B+*+TO+*+%5D+AND+critical pressure mf%3A%5B+*+TO+*+%5D+AND+critical density mf%3A%5B+*+TO+*+%5D&columns=10101010%2C1%2C2%2C5%2C3%2C4%2C6%2C7%2C8%2C9%2C10%2C11%2C12%2C13%2C14%2C15%2C16&data search query=%3Fmn%3DH2O%3Aand%3A%5B+*+TO+*+%5D+AND+critical temperature mf%3A%5B+*+TO+*+%5D+AND+critical pressure mf%3A%5B+*+TO+*+%5D+AND+critical density mf%3A%5B+*+TO+*+%5D&sort=12&sort on=default](https://app-knovel-com.ezproxy1.bath.ac.uk/kn/resources/kt009ZN2P4/kpYCPDCECD/itable/itable?b-q=material%3A+H2O+AND+critical%20temperature_mf%3A%5B+*+TO+*+%5D+AND+critical%20pressure_mf%3A%5B+*+TO+*+%5D+AND+critical%20density_mf%3A%5B+*+TO+*+%5D&columns=10101010%2C1%2C2%2C5%2C3%2C4%2C6%2C7%2C8%2C9%2C10%2C11%2C12%2C13%2C14%2C15%2C16&data_search_query=%3Fmn%3DH2O%3Aand%3A%5B+*+TO+*+%5D+AND+critical%20temperature_mf%3A%5B+*+TO+*+%5D+AND+critical%20pressure_mf%3A%5B+*+TO+*+%5D+AND+critical%20density_mf%3A%5B+*+TO+*+%5D&sort=12&sort_on=default) [Accessed 3 December 2025].